

# WASP-43b

R-1865

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-43\_160203 · created 2026-08-03T10:05:00+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2457421.8217 +/- 0.0011 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.024 +/- 0.057                 |
| Transit depth (Rp/Rs)^2                           | 0.059 +/- 0.27 [%]              |
| Orbital Inclination (inc)                         | 79.9 +/- 2.82                   |
| Airmass coefficient 1 (a1)                        | 0.89 +/- 0.5                    |
| Airmass coefficient 2 (a2)                        | 0.17 +/- 0.55                   |
| Scatter in the residuals of the lightcurve fit is | 56.14 %                         |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 2.76                            |
| Optimal Annulus                                   | 5.75                            |
| Transit Duration (day)                            | 0.028 +/- 0.018                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.024 ± 0.057 vs 0.1594 (deviation 2.07σ) |
| Duration fit vs published | 0.028 d vs 0.0512 d (deviation 1.13σ)     |
| Mid-time O–C              | -1.7 min                                  |
| Depth SNR                 | 0.4                                       |
| Residual scatter          | 56.14 %                                   |

### Light curve

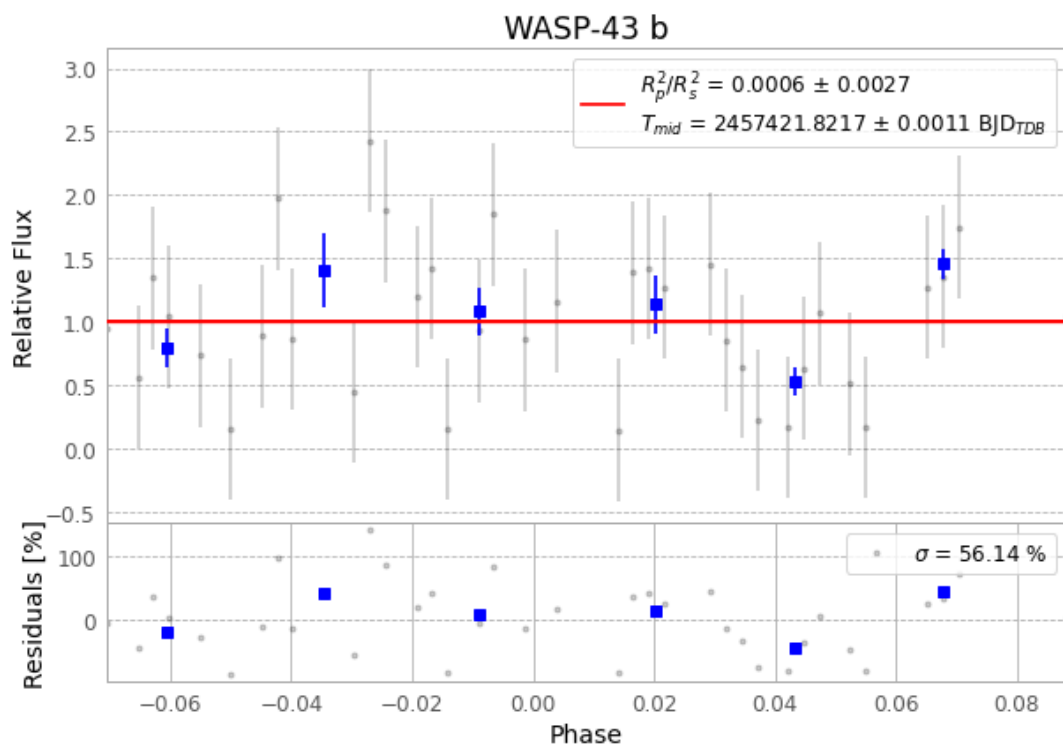


Figure 1 — FinalLightCurve\_WASP-43 b\_2016-02-02.png (SHA-256 6b62191474d9c9dc...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [189, 136], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 df8eaac72fff650d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

64 FITS frames (42 MB), WASP-43160203061212.FITS → WASP-43160203092112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-160203.log (9df4386307e4...), FinalLightCurve\_WASP-43 b\_2016-02-02.png (6b62191474d9...), FinalLightCurve\_WASP-43 b\_2016-02-02.csv (2581b13cb6ad...)

## Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 10:05Z.